$$A=\begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix} \qquad B=\begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix}$$

$$AB=\begin{bmatrix} m_1+m_4-m_5+m_7 & m_3+m_5 \\ m_2+m_4 & m_1+m_3-m_2+m_6 \end{bmatrix}$$

$$m_1=(a_{00}+a_{11})*(b_{00}+b_{11})$$

$$m_2=(a_{10}+a_{11})b_{00}$$

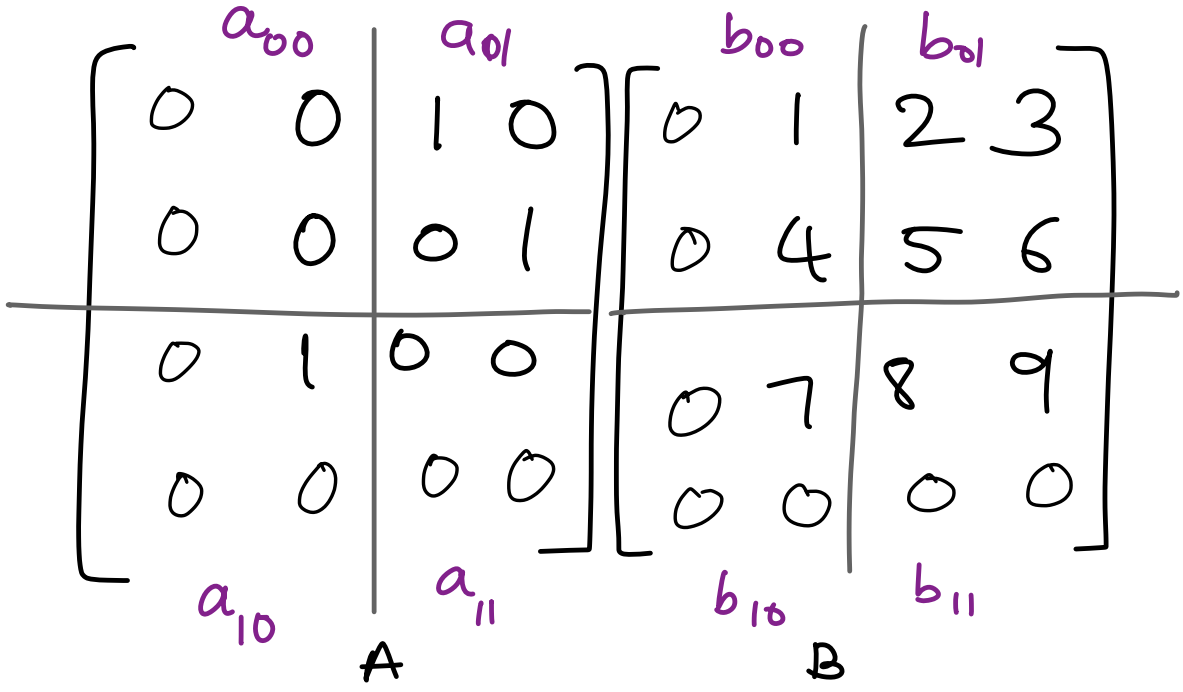
$$m_3=a_{00}(b_{01}-b_{11}) \qquad m_6=(a_{10}-a_{00})*(b_{00}+b_{01})$$

$$m_4=a_{11}(b_{10}-b_{00}) \qquad m_7=(a_{01}-a_{11})*(b_{10}+b_{11})$$

$$m_5=(a_{00}+a_{01})b_{11}$$

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Strassen's Multiplication Illustration:



$$AB = \left[\begin{array}{cc|cc} 0 & 7 & 8 & 9 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 4 & 5 & 6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

(Answer by usual multiplication)

$$m_3 + m_5 = a_{00}(b_{01} - b_{11}) + (a_{00} + a_{01}) b_{11}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}(b_{01} - b_{11}) + \left[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right] \begin{bmatrix} 8 & 9 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 9 \\ 0 & 0 \end{bmatrix}$$

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$$m_2+m_4 = (a_{10}+a_{11})\,b_{00} + a_{11}(b_{10}-b_{00})$$

$$= \left[\begin{pmatrix}0&1\\0&0\end{pmatrix}+\begin{pmatrix}0&0\\0&0\end{pmatrix}\right]\begin{bmatrix}0&1\\0&4\end{bmatrix} + \begin{bmatrix}0&0\\0&0\end{bmatrix}(b_{10}-b_{00})$$

$$= \begin{bmatrix}0&1\\0&0\end{bmatrix}\begin{bmatrix}0&1\\0&4\end{bmatrix}$$

$$= \begin{bmatrix}0&4\\0&0\end{bmatrix}$$ ✓

$m_1 + m_4 - m_5 + m_7$

$= (a_{00} + a_{11}) * (b_{00} + b_{11}) + a_{11}(b_{10} - b_{00}) - (a_{00} + a_{01}) b_{11}$

$+ (a_{01} - a_{11}) * (b_{10} + b_{11})$

$= \left(\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 00 \\ 00 \end{bmatrix}\right) * \left(\begin{bmatrix} 0 & 1 \\ 04 \end{bmatrix} + \begin{bmatrix} 8 & 9 \\ 00 \end{bmatrix}\right) + \begin{bmatrix} 00 \\ 00 \end{bmatrix} - \left(\left(\begin{bmatrix} 00 \\ 00 \end{bmatrix} + \begin{bmatrix} 10 \\ 01 \end{bmatrix}\right)\begin{pmatrix} 8 & 9 \\ 0 & 0 \end{pmatrix}\right)$

$+ \left(\left(\begin{bmatrix} 10 \\ 01 \end{bmatrix} - \begin{bmatrix} 00 \\ 00 \end{bmatrix}\right) * \left(\begin{bmatrix} 0 \\ 00 \end{bmatrix} + \begin{bmatrix} 8 & 9 \\ 0 & 0 \end{bmatrix}\right)\right)$

$= \begin{bmatrix} 00 \\ 00 \end{bmatrix} + \begin{bmatrix} 00 \\ 00 \end{bmatrix} - \begin{bmatrix} 8 & 9 \\ 0 & 0 \end{bmatrix} + \left(\begin{bmatrix} 10 \\ 01 \end{bmatrix} * \begin{bmatrix} 8 & 16 \\ 0 & 0 \end{bmatrix}\right)$

$= \begin{bmatrix} 0 & 7 \\ 0 & 0 \end{bmatrix}$ ✓

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$m_1 + m_3 - m_2 + m_6$

$= (a_{00}+a_{11})*(b_{00}+b_{11}) + a_{00}(b_{01}-b_{11}) - (a_{10}+a_{11})b_{00}$

$+ (a_{10}-a_{00})*(b_{00}+b_{01})$

$= \left[\begin{pmatrix}0&0\\0&0\end{pmatrix}+\begin{pmatrix}0&0\\0&0\end{pmatrix}\right]*\left[b_{00}+b_{11}\right] + \begin{pmatrix}0&0\\0&0\end{pmatrix} - \left[\begin{pmatrix}0&1\\0&0\end{pmatrix}+\begin{pmatrix}0&0\\0&0\end{pmatrix}\right]\begin{bmatrix}0&1\\0&4\end{bmatrix}$

$+ \left[\begin{pmatrix}0&1\\0&0\end{pmatrix}-\begin{pmatrix}0&0\\0&0\end{pmatrix}\right]*\left[\begin{pmatrix}0&1\\0&4\end{pmatrix}+\begin{pmatrix}2&3\\5&6\end{pmatrix}\right]$

$= \begin{bmatrix}0&0\\0&0\end{bmatrix} + \begin{bmatrix}0&0\\0&0\end{bmatrix} - \begin{bmatrix}0&1\\0&0\end{bmatrix}\begin{bmatrix}0&1\\0&4\end{bmatrix} + \begin{bmatrix}0&1\\0&0\end{bmatrix}\begin{bmatrix}2&4\\5&10\end{bmatrix}$

$= -\begin{bmatrix}0&4\\0&0\end{bmatrix} + \begin{bmatrix}5&10\\0&0\end{bmatrix} = \begin{bmatrix}5&6\\0&0\end{bmatrix}$ ✓

Dynamic Programming:

Idea: "Bottom-up approach."

Fibonacci Sequence:

$0, 1, 2, 3, 5, 8, 13, \ldots$

$$F_n = F_{n-1} + F_{n-2}, \quad n \geq 2.$$

$$F_0 = 0, \quad F_1 = 1$$

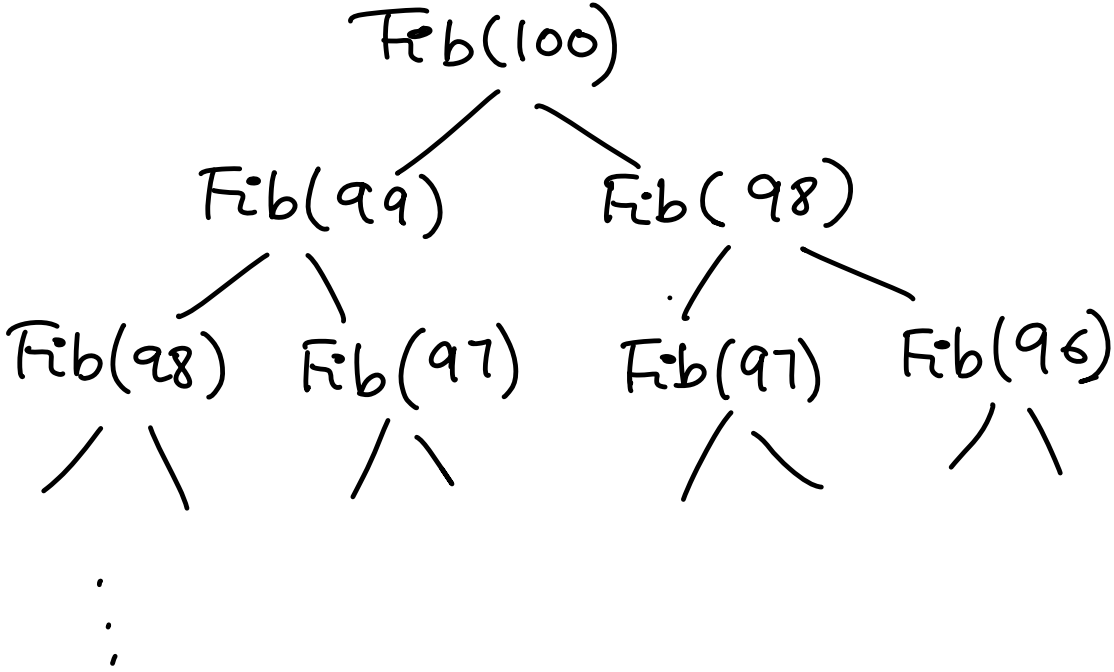
Fib(n):

If $n = 0$ return 0

$n = 1$ return 1

else

return $\left(Fib(n-1) + Fib(n-2)\right)$.


Fib(100)
Fib(99)
Fib(98)
Fib(98)
Fib(97)
Fib(97)
Fib(96)

Fib(n) Time Complexity:

$$T(n) = T(n-1) + T(n-2)$$

Solve $T_n = T_{n-1} + T_{n-2}$

$$T_{n+2} = T_{n-1} + T_n$$

$$T_{n+2} - T_{n-1} - T_n = 0$$

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$$m^2 - m - 1 = 0$$

$ax^2+bx+c=0$

$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$$m=\frac{-(-1)\pm\sqrt{(-1)^2-4(1)(-1)}}{2(1)}$$

$$=\frac{1\pm\sqrt{1+4}}{2}$$

$$=\frac{1+\sqrt{5}}{2},\quad \frac{1-\sqrt{5}}{2}$$

m_1 $\quad$ m_2

$$\Rightarrow T_n=c_1 m_1^n+c_2 m_2^n$$

$$T_n = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

Using $T_0 = 0$, $T_1 = 1$. to find c_1 & c_2.

OR

$$T(n) = T(n-1) + T(n-2)$$

$$\geq T(n-2) + T(n-2)$$

$$= 2T(n-2)$$

replace $n \to n-2$

$$\geq 2\left(2T(n-4)\right)$$

$$= 2^2\left(T(n-4)\right)$$

$$\geq 2^3 T(n-6)$$

$$\vdots$$

$\geq 2^k T(1)$

$n-2k=1$

$n-1=2k$

$k=\frac{n-1}{2}$

$= 2^{\frac{n-1}{2}} \cdot 1 \in O(2^n)$

(Exponential)

$T(n) \geq 2^{\frac{n-1}{2}}$

$T(n) \in O(2^n)$

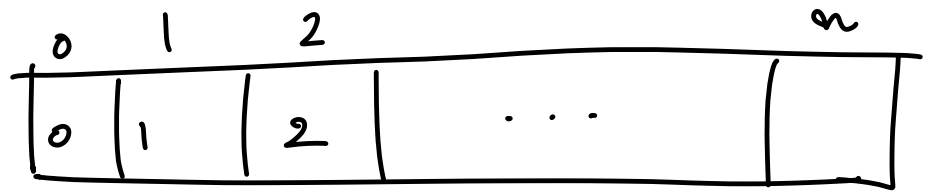
Fib(n) is exponential algorithm.

Solution to bring to linear

① "memoization."
(Store the previous information)

② Not to use Recursive Relation redundantly.

Linear:



F(n).

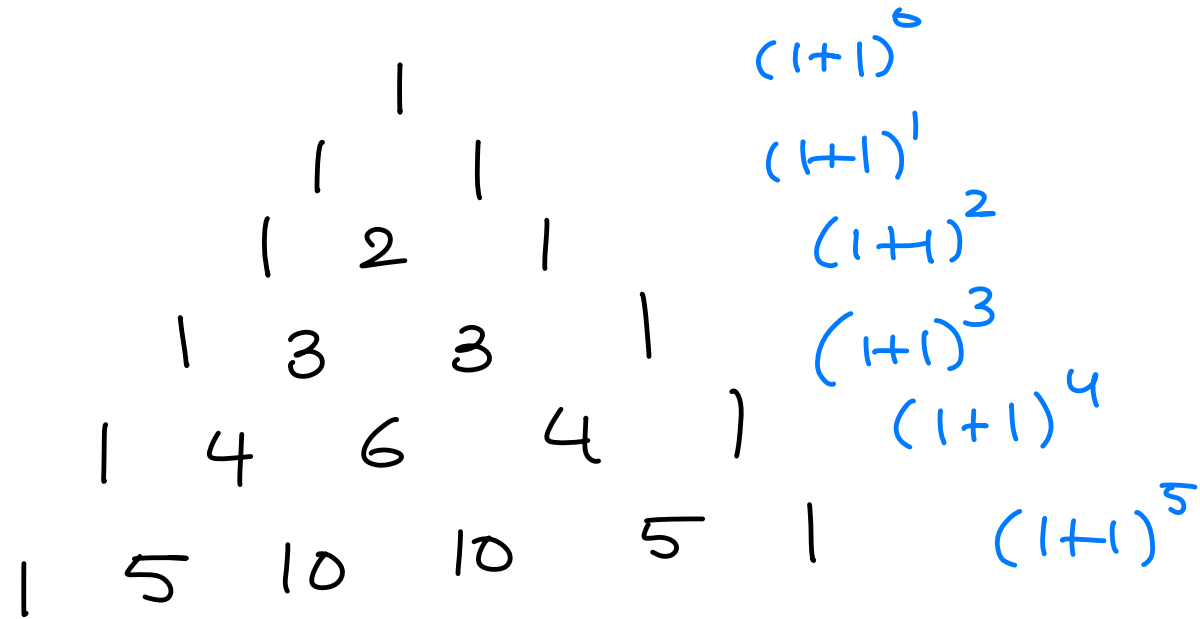
$F(0) := 0$; $F(1) := 1$;

For $i = 2$ to n

$$F(i) = F(i-1) + F(i-2).$$

$$T(n) = \sum_{i=2}^{n} 1 = n - 1 \in \Theta(n).$$

Pascal's Triangle :

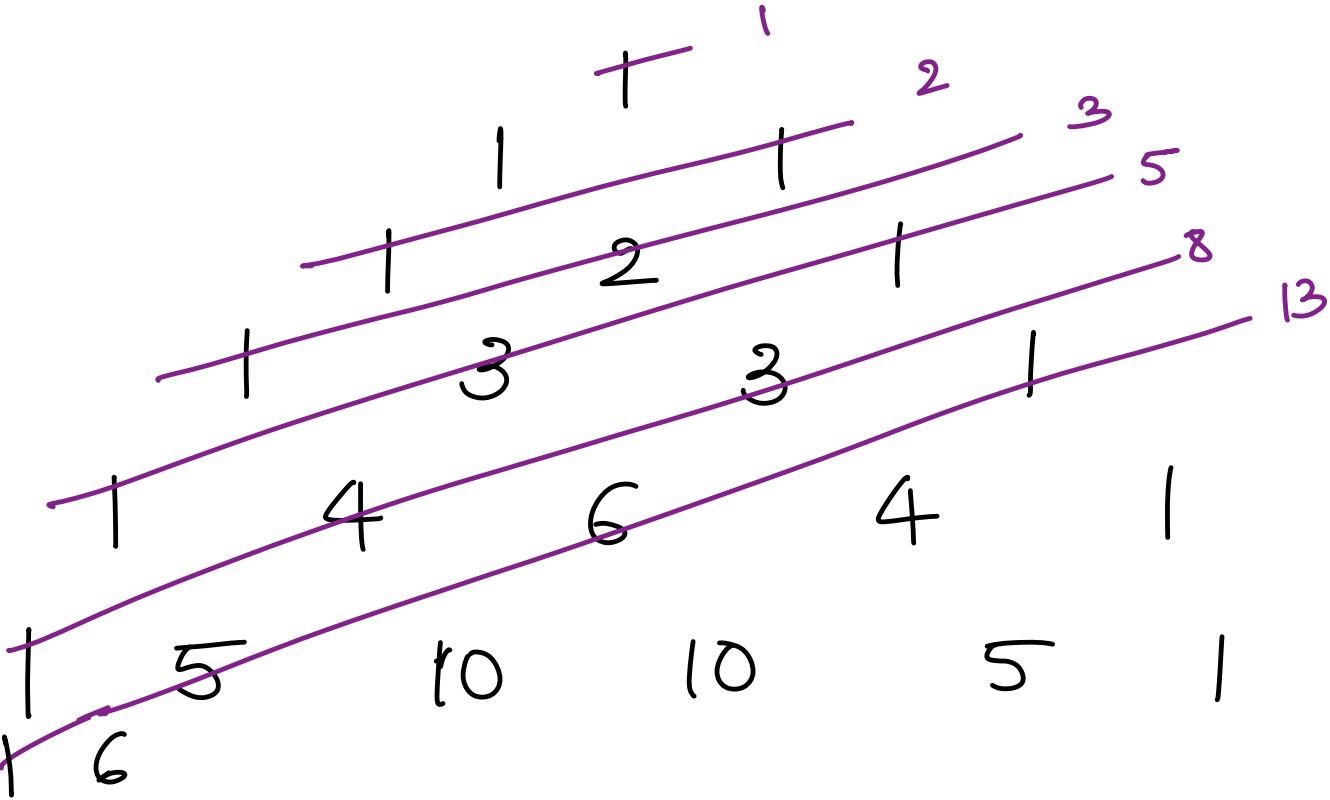


$$(x+y)^5$$

$$= (1)x^5 + (5)x^4y + (10)x^3y^2$$

$$+ (10)x^2y^3 + (5)xy^4 + (1)y^5$$

Fibonacci numbers in Pascal's Δle:



Top Down Approach: (Recurrence Relation)

$$^{n}C_{r} = {}^{n-1}C_{r} + {}^{n-1}C_{r-1}$$

$$P(n,r) = P(n-1,r) + P(n-1,r-1)$$

Base: $P(0,0) = 1$

$P(1,1) = 1$

$P(1,2) = 1$

$P(n,r)$:

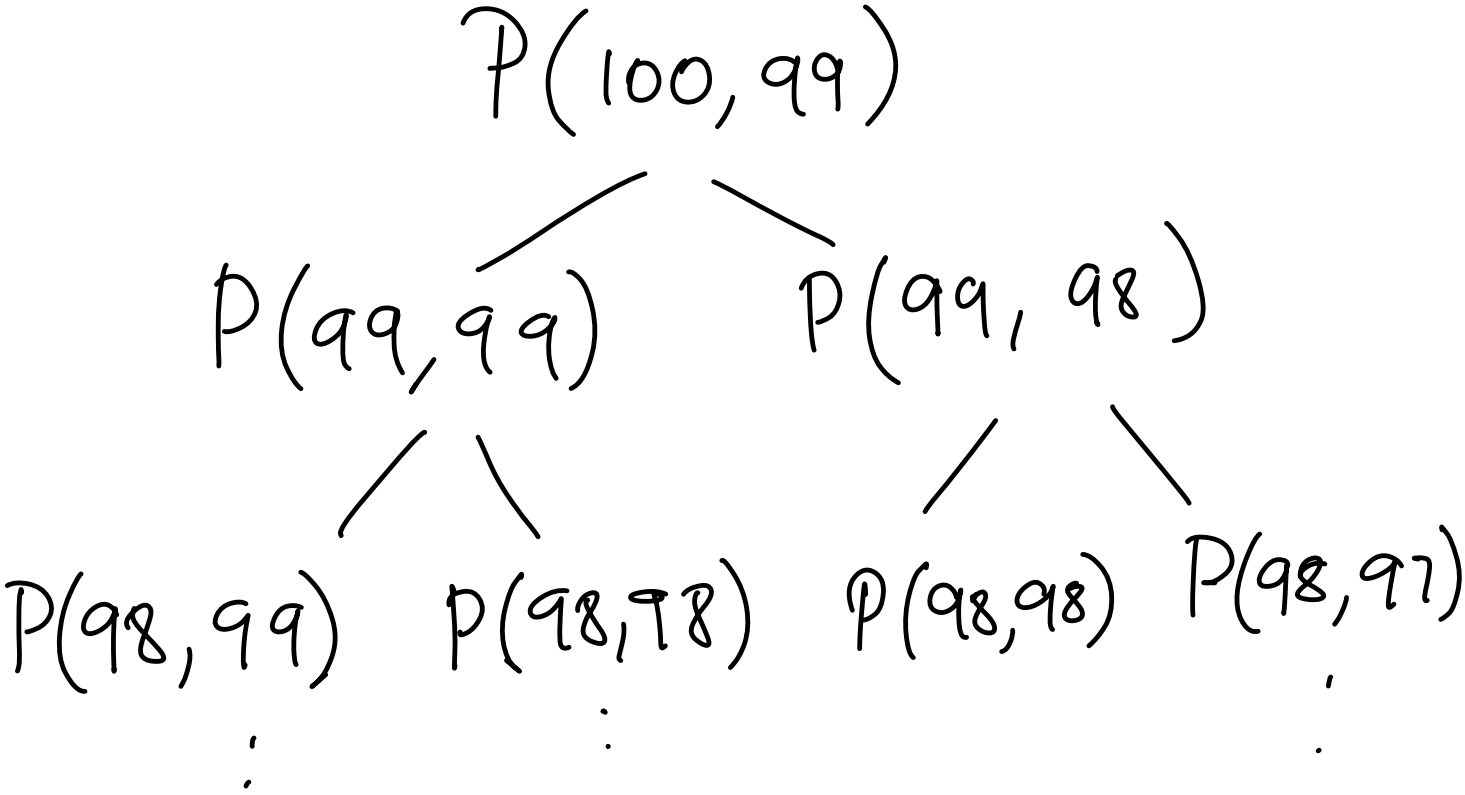
If $(n = 0 \text{ or } 1)$ and

$(r = 0, 1, 2)$ return 1

else

$$P(n,r) = P(n-1, r) + P(n-1, r-1).$$

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P(100, 99)
P(99, 99)
P(99, 98)
P(98, 99)
P(98, 98)
P(98, 98)
P(98, 97)

Time Complexity:

Let size $= n + n = N$

$$T(size) = T(size-1) + T(size-2)$$

$$T(N) = T(N-1) + T(N-2)$$

$$T(N) \geq 2^{\frac{N-1}{2}} \quad \text{(Exponential)}$$